

Convergence Tests



The ratio and comparison tests

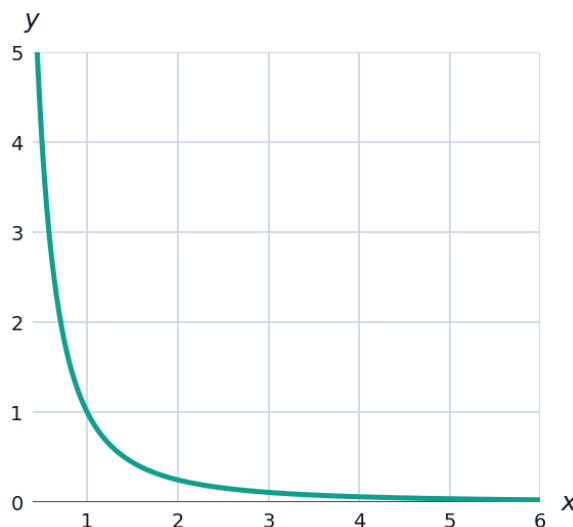
- 1 Use the ratio test to determine whether $\sum_{n=1}^{\infty} \frac{n}{2^n}$ converges.
- 2 Use the ratio test on $\sum_{n=1}^{\infty} \frac{n!}{n^n}$.
- 3 Use the comparison test to determine whether $\sum_{n=1}^{\infty} \frac{1}{n^2+3}$ converges, by comparing to $\sum \frac{1}{n^2}$.

Alternating series and the p-series test

- 4 Determine whether each p -series converges or diverges:
a $\sum_{n=1}^{\infty} \frac{1}{n^3}$ **b** $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$
- 5 Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ converges, and classify it as absolutely convergent, conditionally convergent, or divergent.

The integral test

- 6 Use the integral test to determine whether $\sum_{n=1}^{\infty} \frac{1}{n \ln n}$ (starting at $n = 2$) converges or diverges, using $\int_2^{\infty} \frac{1}{x \ln x} dx$.
- 7 Use the integral test to confirm that $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges, using $\int_1^{\infty} \frac{1}{x^2} dx$.



Choosing the right test

- 8 For each series, name the most efficient test to apply (ratio, comparison, p -series, or alternating series), without fully evaluating:

a $\sum \frac{2^n}{n!}$

b $\sum \frac{(-1)^n}{\sqrt{n}}$

c $\sum \frac{1}{n^4}$

The limit comparison test

- 9 Use the limit comparison test to determine whether $\sum_{n=1}^{\infty} \frac{n+1}{n^3+2}$ converges, comparing to $\sum \frac{1}{n^2}$.
- 10 Use the limit comparison test on $\sum_{n=1}^{\infty} \frac{3n^2-1}{n^4+2n}$, comparing to $\sum \frac{1}{n^2}$.
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The root test

- 11 Use the root test to determine whether $\sum_{n=1}^{\infty} \left(\frac{n}{2n+1}\right)^n$ converges.
- 12 Use the root test on $\sum_{n=1}^{\infty} \frac{3^n}{n^n}$.
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Absolute versus conditional convergence

- 13 Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$ is absolutely convergent, conditionally convergent, or divergent.
- 14 Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$ is absolutely convergent, conditionally convergent, or divergent.
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Estimating the sum of an alternating series

- 15 The alternating series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$ converges by the alternating series test. Use the fact that the error after n terms is bounded by the first omitted term to estimate how many terms are needed for the partial sum to be within 0.001 of the true sum.
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A combined convergence-test challenge

- 16 Determine whether $\sum_{n=1}^{\infty} \frac{n^2}{3^n}$ converges, using the ratio test.
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- 17 Determine whether $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$ converges, using the integral test with $u = \ln x$.
- 18 A student claims that since $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$, the series $\sum_{n=1}^{\infty} \frac{1}{n}$ must converge. Explain what is wrong with this reasoning, and state the correct necessary (but not sufficient) condition that the n th-term test actually provides, using the harmonic series as a counterexample.

A multi-step convergence investigation

- 19 For the series $\sum_{n=1}^{\infty} \frac{n^3 + 2n}{2n^5 - n + 1}$, first find the n th-term limit to check the necessary condition, then use the limit comparison test with $\sum_{n=1}^{\infty} \frac{1}{n^2}$ to determine whether the series converges.
- 20 A quality-control engineer models cumulative defect probability using $\sum_{n=1}^{\infty} \frac{1}{n \cdot 2^n}$. Use the ratio test to confirm this series converges, and briefly explain (in one sentence) why a convergent bound like this is useful for guaranteeing that a manufacturing process's total defect risk stays finite over infinitely many production runs.